Cassirer's Philosophy of Chemistry

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From Duhem to Bachelard, reflections on chemistry played a central role in French philosophy of science, whereas German-speaking philosophy of science, from Kant to the Vienna Circle, remained predominantly oriented toward physics. This paper presents Cassirer as an overlooked exception to this tendency. In his 1910 *Substanzbegriff und Funktionsbegriff*, he engaged closely with contemporary debates on chemistry and regularly returned to the subject in his subsequent epistemological writings. Despite growing scholarly interest in his historiography and philosophy of mathematics and physics, his reflections on chemistry have received no attention. By addressing this gap, the paper illuminates overlooked aspects of Cassirer's philosophy of the natural sciences and recovers a neglected episode in the early history of the philosophy of chemistry.

Introduction

On September 30, 1909, during a visit to Paris, Hermann Cohen, the founder of the so-called Marburg school of neo-Kantianism, sent Émile Meyerson a postcard asking whether he would be available to meet the following afternoon. A trained chemist without an academic position, Meyerson had just published his first philosophical book, *Identité et réalité* 1908, which, at nearly fifty years of age, brought him almost overnight prominence in Parisian intellectual circles (Fruteau de Laclos 2009). In the same postcard, Cohen noted that one of his former students, Ernst Cassirer, was also in Paris and would join them for the meeting. A few weeks later, Meyerson (Meyerson to Cassirer, Nov. 2, 1909[1])[2] received a copy of *Das Erkenntnisproblem* (Cassirer 1906, 1907). Into a rich tapestry of historical detail, the book famously weaves the thread of a linear historical development of modern science from substance-concepts to function-concepts, an opposition that Cassirer was in the process of developing in his systematic monograph *Substanzbegriff und Funktionsbegriff* (Cassirer 1910).

By late December 1909, Meyerson had contacted Xavier Léon, the editor of the *Revue de métaphysique et de morale*, with the intention of drafting a 'critical review' of Cassirer's work (Meyerson to Cassirer, Dec. 13, 1909[3]). From the outset, Cassirer welcomed Meyerson's proposal with great enthusiasm (Cassirer to Meyerson, Dec. 21, 1909[4]). Cassirer was clearly eager to hear his reaction before starting work on the second

[1] ECN, Vol. 18, Doc. 121.

[2] The numbering of the Meyerson–Cassirer letters follows Vol. 18 of the edition of Cassirer's *Nachlass*. However, the letters are not transcribed. Scans of the originals are available online at Hamburger Kulturgut Digital; last consulted on November 3, 2026.

[3] Vol. 18, Doc. 132.

[4] Vol. 18, Doc. 133.

edition of *Das Erkenntnisproblem* (Cassirer to Meyerson, Dec. 21, 1909[5]; Cassirer to Meyerson, Jan. 1, 1910[6]). Meyerson wrote to Cassirer a few months later, in April 1910, after completing the 'second reading' of *Das Erkenntnisproblem*, announcing a visit to Berlin in late April and early May 1910 (Meyerson to Cassirer, Apr. 18, 1910[7]). Their correspondence sheds no light on what they discussed. Nevertheless, it seems plausible that Meyerson anticipated the criticism that would later appear in his published review (Meyerson 1911). According to Meyerson, *Das Erkenntnisproblem* completely neglects the role of chemistry in the historical development of the 'problem of knowledge' in early modern science. Cassirer's exclusive focus on the history of physics, Meyerson argued, led him to overstate the importance of function-concepts at the expense of the substance-concept.

At that time, Cassirer was already in the final stages of drafting *Substanzbegriff und Funktionsbegriff* (Cassirer 1910), which was completed in July 1910. Meyerson received at least part of the galley proofs shortly thereafter (Meyerson to Cassirer, Jul. 17, 1910[8]), but the complete book only much later, when the review was already in press (Meyerson to Cassirer, Oct. 21, 1910[9]). Had he had access to the book, he would have realized that, far from neglecting chemistry, Cassirer had devoted a rather well-informed twenty-page section to the subject (270–92/203–20).[10] Cassirer's choice is by no means obvious. Even the word 'chemistry' is absent from Paul Natorp's (1910) contemporaneous 'Marburg' monograph on the foundations of the 'exact sciences.' For Cassirer, by contrast, chemistry has a peculiar epistemological status with respect to physics. In both cases, scientific concept formation proceeds from substance-concepts to function-concepts; however, while physics foregrounds the movement toward increasingly general laws, only chemistry brings into view the complementary movement toward increasingly determinate individuality (Cassirer 1910, 270–72/203–5; 289–93/217–21).

Meyerson's (1911) review finally appeared in the January issue of the 1911 volume of the *Revue de métaphysique*. Cassirer read it immediately and conceded that, from a *historiographical* perspective, *Das Erkenntnisproblem* had devoted less attention to the history of chemistry than it deserved; indeed, he added to the second edition an entire new section addressing Meyerson's complaints. However, from a *systematic* point of view, Cassirer integrated this material into a conceptual framework very different from the one Meyerson had in mind. On Meyerson's account, the history of chemistry unmistakably shows that scientific explanation proceeds by progressively eliminating distinctions until matter is reduced to an undifferentiated unity (Meyerson 1908, ch. VII). For Cassirer, by contrast, the history of chemistry shows more clearly than that of physics that the 'generality' of scientific concept formation aims at increasing individual determinacy. The discussion of chemistry in section 4.VIII of *Substanzbegriff und Funktionsbegriff* is not merely an appendix to the preceding sections on physics. Rather, it lays the groundwork for section 4.IX, in which Cassirer draws on chemistry to challenge Heinrich Rickert's (1902) distinction between the 'generalizing' natural sciences and the 'individualizing' historical sciences. The case of chemistry thus enables Cassirer to bring together the arguments developed in the first part of the book and

[5] ECN, Vol. 18, Doc. 133.
[6] Vol. 18, Doc. 134.
[7] Vol. 18, Doc. 141.
[8] Vol. 18, Doc. 148.
[9] Vol. 18, Doc. 157.
[10] The number following the slash indicates the corresponding page in the English translation given alongside the original edition in the bibliography.

situate them within the broader debate between the two major neo-Kantian schools of the period.

As has been noted, reflections on chemistry, from Duhem to Meyerson, played a central role in early French philosophy of science (Bensaude-Vincent 2009). By contrast, German philosophy, perhaps following Kant's own denial of chemistry's status as a 'science,' had always remained physics-oriented, despite the fact that chemistry had become one of the country's most successful scientific disciplines during the nineteenth century. In this context, Cassirer's reflections on chemistry are all the more remarkable. However, despite the increasing attention given to Cassirer's historiography and philosophy of mathematics (Reck 2020) and physics (Ryckman 2005), these reflections have received no scholarly attention. Against this background, the present paper examines Cassirer's engagement with chemistry, reconstructing its historical context (section 1) and exploring its role in his confrontation with Rickert (section 2) and Meyerson (section 3). In doing so, the paper aims to shed light on overlooked aspects of Cassirer's philosophy, particularly his attitude toward atomism, while also recovering a neglected episode in the pre-history of the philosophy of chemistry (Brakel 2012). Whereas Meyerson's interest in chemistry left a lasting legacy in France, extending from Hélène Metzger to Gaston Bachelard, Cassirer's philosophy of chemistry appears to have had little resonance, not only within neo-Kantianism but also in German-speaking philosophy of science, which remained largely focused on physics.

1 Chemistry in *Substanzbegriff und Funktionsbegriff*

Cassirer referred to chemistry throughout *Substanzbegriff und Funktionsbegriff*, but devoted section VIII of the fourth chapter, on concept formation (*Begriffsbildung*) in the natural sciences, specifically to chemical concept formation. Chemistry had initially developed as a purely 'descriptive science,' much like zoology or botany. However, by the time Cassirer was writing, the emergence of physical chemistry in the late nineteenth century had firmly established chemistry among the 'constructive sciences' (see Nernst 1896). In particular, chemists increasingly turned to thermodynamics to investigate chemical processes, analyzing reaction rates and equilibrium conditions in terms of macroscopic variables such as temperature, pressure, and energy. As Cassirer points out, from this point of view, chemistry's "fundamental laws, as for example the law of phases (*Phasenregel*) of Gibbs or the law of chemical masses (*Massenwirkung*), belong to the same purely mathematical type as the propositions of theoretical physics" (Cassirer 1910, 271/204). The difference between physics and chemistry thus lies not in their status as 'exact sciences,' but in the historical paths by which they attained that status.

Physics had the advantage of dealing with relatively simple systems, such as planetary motion, in which bodies could be treated as 'structureless' mass points. It could therefore replace the thing-concept (*Dingbegriff*) with the concept of law (*Gesetzesbegriff*) at an early stage. Chemistry, by contrast, had to begin with the problem of the individual thing (*Einzelding*), that is, with the specific properties of elements and compounds in all their complexity (271/204). Cassirer therefore asks whether chemistry requires a different form of concept formation from that of physics. As one might expect, Cassirer argues that this question can be answered only "if we follow the concrete historical development of chemical doctrines themselves" (272/204). Given the systematic aims of the book, Cassirer does not attempt to trace the history

of chemistry in detail, but seeks instead to identify "the great logical lines of direction" along which it developed (Cassirer 1910, 272/204). Cassirer does not appear to draw directly on the original sources, relying instead on Wurtz (1879), Meyer (1884), Berthelot (1885), Ladenburg (1902), Duhem (1902), and Ostwald (1906) as his principal historical accounts of the development of chemistry. Cassirer reorganizes this material to fit his own philosophical narrative.

1.1 From Definite to Multiple Proportions

As one might expect, Cassirer emphasizes that concept formation in early chemistry followed the logical presuppositions of the Aristotelian system, which moves from the genus-concept (*Gattungsbegriff*) to its metaphysical counterpart, the substance-concept (*Substanzbegriff*). Medieval natural science, particularly chemistry and alchemy, can be understood only in light of the Aristotelian model, which treats qualities as independent entities that can, at least in principle, be transferred from one body to another (Cassirer 1910, 204/154). The hypothesis of phlogiston shows how slowly this conception was abandoned in chemistry (205/155[11]): combustibility, a property that we seem to perceive directly in various bodies, was conceived as a distinct substance admixed with those bodies: "and from this assumption, the whole structure of chemistry before Lavoisier follows with inner necessity" (205/155). In a footnote, Cassirer refers to the "excellent presentation" in Meyerson's (1908) book, which assigns Lavoisier's establishment of the law of 'conservation of mass' a pivotal role in the quantitative turn in the history of chemistry. Cassirer may well have added this reference at a late stage of revision. Indeed, his account immediately diverges from Meyerson's by assigning a central role instead to the German chemist Jeremias Benjamin Richter, whom Meyerson, by contrast, does not discuss (see Duhem 1902, III; Ladenburg 1902, lect. 4; Wurtz 1879, I.I; Ostwald 1906, 44–49).

In Cassirer's view, it was Richter's 'law of definite or constant proportions' that constituted "the starting-point of modern chemical *theory*" (Cassirer 1910, 273/205) and marked a decisive break with the alchemical tradition. Richter observed that definite masses of different bases were required to neutralize a fixed mass of a given acid, and that the ratios between these masses remained constant for other acids. His discovery thus established not merely the *definite* proportions characteristic of chemical compounds, as opposed to the *variable* proportions of physical mixtures, but also fixed quantitative relations between different chemical combinations. According to Cassirer, as in physics, "[t]he demand for strict numerical determination here also forms the decisive turning point" (272/205). What matters for Cassirer is therefore not merely the conservation (*Beharrung*) of a quantity, as in Lavoisier, but the establishment of a coordination (*Zuordnung*) between distinct ordered sequences of numbers.

Indeed, these laws could be established by measuring the masses of substances involved in chemical reactions "entirely independently of any conception of the constitution of matter, and in particular independently of the atomic hypothesis" (273/205). Richter determined the masses of different bases neutralized by a given mass of sulfuric acid, hydrofluoric acid, and other acids, arranging them into numerical series. Conversely, he determined the masses of different acids required to neutralize a given mass of a base. Following Duhem (1902, 71), Cassirer introduces a series of acids and a series of bases:

[11] Translation modified.

$$A_1, A_2, A_3, \ldots \qquad B_1, B_2, B_3, \ldots ,$$

and establishes a correspondence between these substances and two series of characteristic numbers:

$$m_1, m_2, m_3, \ldots \qquad n_1, n_2, n_3, \ldots ,$$

so that every acid A_i is assigned a characteristic number m_i, and every base B_j a characteristic number n_j. Suppose sulfuric acid is assigned the number $m_1 = 100$. Since approximately 82 grams of sodium hydroxide neutralize 100 grams of sulfuric acid, sodium hydroxide is assigned $n_1 = 82$. Since approximately 74 grams of hydrochloric acid neutralize the same mass of sodium hydroxide, hydrochloric acid is assigned $m_2 = 74$. Each substance is thus characterized by a number expressing its quantitative combining relations. Once m_p and n_q are known, the combining ratio of any acid A_p and base B_q is uniquely determined, without further measurements: "The manner, in which an element of the first series combines with an element of the second, is definitely [*eindeutig*] determined by these numbers; the two weights, according to which an acid A_p combines with any base B_q, are related according to the corresponding numerical values m_p and n_q" (Cassirer 1910, 273/205).

As the conventional rational reconstruction goes, the generalization of Richter's scheme encountered a difficulty: the same two elements can form more than one compound, each with its own definite composition. In other words, they do not always combine in a single fixed ratio. Carbon and oxygen, for example, form both carbon monoxide and carbon dioxide. In the former, 12g of carbon combine with 16g of oxygen; in the latter, 12g of carbon combine with 32g of oxygen. A single fixed combining ratio cannot account for this variation. John Dalton provided a solution by formulating the 'law of multiple proportions.' Instead of abandoning the idea of characteristic masses (p_i), he retained them as fundamental proportional numbers and introduced integer multipliers (λ, μ, ν). The mass ratio of any compound is then no longer simply $p_1 : p_2$, but

$$\lambda p_1 : \mu p_2 : \nu p_3 : \cdots$$

where p_i is the characteristic proportional mass of an element (e.g., $p_C = 12$, $p_O = 16$, using modern values), and λ, μ, ν are positive integers $(1, 2, 3, \ldots)$. In carbon monoxide, the mass ratio is $1{\cdot}12 : 1{\cdot}16$, so that $\lambda = 1$ and $\mu = 1$; in carbon dioxide, it is $1{\cdot}12 : 2{\cdot}16$, so that $\lambda = 1$ and $\mu = 2$. Berzelius's chemical notation subsequently provided a symbolic representation of these combining proportions. On a purely quantitative reading, a formula such as

$$\mathrm{H_2O} \quad \text{or} \quad \mathrm{CO_2}$$

can be interpreted as an algebraic shorthand for the corresponding integer multipliers. The letters designate elements associated with characteristic proportional masses (p_1, p_2), and the subscripts indicate their multipliers (λ, μ).

1.2 From Equivalent Weights to Atomic Weights

In this sense, Cassirer could point out that the compositional formula has a purely macroscopic quantitative content, independent of its interpretation in terms of atoms.

The goal of the 'law of constant combining numbers' (*Verbindungszahlen*) was initially to establish a characteristic equivalent number for each element (*Äquivalenzzahl*), such that whenever two or more elements enter into a compound, their masses stand to one another "as whole multiples of these numbers" (Cassirer 1910, 274/206). In principle, these ratios can be established by measuring the macroscopic masses of substances involved in chemical reactions. In Dalton's work, however, this rule of 'multiple proportions' becomes associated with a particular interpretation: "The concept of combination-weight [*Verbindungsgewichts*] is transformed into that of atomic weight [*Atomgewichts*]" (274/206). Although the law does not itself require matter to be discontinuous, the occurrence of simple integer ratios receives a natural explanation if chemical elements consist of discrete units with fixed masses. Dalton interpreted such regularities by assuming that substances consist of indivisible atoms, each element possessing atoms of a characteristic mass, with hydrogen = 1 as the unit standard. Different elements possess atoms of different masses, while all atoms of the same element possess the same constant mass: "In place of the empirically gained proportion-numbers (*Proportionalzahlen*) of the individual bodies, assertions appear regarding an essential property of their ultimate constitutive parts" (274/206).

In the rational reconstruction that Cassirer broadly follows, Dalton's hypothesis gives rise to a fundamental difficulty. The empirical facts do not uniquely determine the relative atomic weights. Suppose hydrogen is taken as the unit: the same experimental composition can be represented using different atomic weights for oxygen, provided that the chemical formulas are altered accordingly. Thus, the experimental facts of composition alone cannot decide between, for example, $O = 16$ and $O = 8$. The same applies to sulfur, for which values such as $S = 8, \quad S = 16, \quad S = 32, \ldots$ remain possible in the absence of further constraints (274–75/206–7). In principle, chemists could have accepted this ambiguity as unavoidable; instead, they sought additional, independent relations that would allow them to determine the atomic weights unambiguously. Following Meyer's (1884) account, Cassirer shows how the *univocal determination* of relative atomic weights emerges progressively, not from a single measurement, but from the mutual confirmation and correction of results obtained through independent physical laws. He mentions three particularly important ones: Avogadro's rule,[12] relating molecular weights to gas volumes and vapor densities; the Dulong–Petit law, relating atomic weights to specific heats[13]; and Mitscherlich's law of isomorphism, relating crystalline form to chemical composition and structure[14] (Cassirer 1910, 275/207). No single rule can univocally determine all the relative atomic weights. Rather, these weights become increasingly determinate as several initially distinct series of phenomena are brought into mutual agreement: "It is only the totality of all these different points of view, mutually confirming and correcting each other, that finally after many experiments gives a unitary table of atomic weights, and thus lays the basis of a definite system of chemical formulae" (275/207).

Here the conditions of "scientific *production*" must be distinguished from those of "critical *reflection*" (279/210). A modern chemist might maintain that "[t]he objective

[12]Equal volumes of gases at the same pressure and temperature contain equal numbers of molecules ($M \propto d$, where M is molar mass and d is gas density).

[13]The product of an element's atomic weight and its specific heat is approximately constant ($A \cdot c \approx 6.0$–6.4, with specific heat expressed in calories per gram per kelvin), within the law's range of applicability.

[14]Isomorphism provides evidence of chemical analogy and can therefore help establish corresponding chemical formulas.

existence of the different types of atoms is presupposed; it is only necessary to discover their *properties*, and to define them more exactly" (Cassirer 1910, 275/207). Although scientists may take atoms as their starting point, Cassirer, following Duhem and Ostwald, insists that the empirical basis of atomic theory reveals the reverse. Inquiry begins with experimental laws (conservation of mass and definite, reciprocal, and multiple proportions) that establish invariant mass relations in chemical reactions and distinguish compounds from mixtures. The concept of 'combining weight' is then reinterpreted as the 'atomic weight' of discrete constituents whose combinations explain the observed whole-number ratios. Determining atomic weights, however, requires coordinating further independent empirical relations. Thus, "[i]t becomes clear that the atom is never the given *starting-point*, but always only the *goal* of our scientific statements" (276/208).

As expected, Cassirer rejects Wurtz's naive atomism. It is worth noting, however, that he does not embrace Duhem's militant anti-atomism either. For Cassirer, the alternative itself is ultimately misleading. The decisive question concerns not "metaphysical assertions regarding the *existence of atoms*", but "the characteristic *logical function of the atom-concept*" (276–77/208[15]). If one confines oneself strictly to observational data, one might simply *accept* as an empirical fact that elements combine in several distinct ratios. Daltonian atomism, by contrast, introduced the problem of *resolving* the ambiguity by establishing a unique system of atomic weights, a task that ultimately required recourse to independent physical laws. Although chemistry initially appears to investigate the properties of atoms as *bearers* of constant atomic weights, closer examination reveals that it employs the *series* of atomic weights to establish univocal correspondences between otherwise unrelated domains of experience. The Avogadro–Cannizzaro method, for example, establishes a relation between the vapor densities of gaseous substances,

$$\rho_1, \rho_2, \ldots, \rho_n \,,$$

and their molecular weights. Since the relative molecular weights of gases are proportional to their densities at the same temperature and pressure, the atomic weights of their constituent elements,

$$a_1, a_2, \ldots, a_n \,,$$

can be determined by comparing the molecular weights and chemical compositions of different compounds. The Dulong–Petit law, in turn, establishes a correspondence between these same atomic weights and the specific heats of solid elements,

$$b_1, b_2, \ldots, b_n \,,$$

such that the product of an element's atomic weight and its specific heat is approximately constant. The same series of atomic weights thus mediates between two apparently unrelated empirical domains: the behavior of gases and the thermal properties of solids (276–77/208).

The question of the reality of the 'atom' is thus displaced by the question of the heuristic role of the atom-concept in establishing univocal correspondences between otherwise unrelated series: "The atom functions here as the conceived unitary center of a

[15]Emphasis mine.

system of coördinates, in which we conceive all assertions concerning the various groups of chemical properties arranged" (Cassirer 1910, 277/208). The initially heterogeneous numerical series are brought into univocal coordination (*eindeutige Zuordnung*) by relating each of them to the mediating series of atomic weights. The atom with its determinate atomic weight is not simply an underlying 'bearer' of independently discovered properties. Rather, the series of atomic weights serves as the "*common series*" through which different numerical series are related to one another: "In truth, we are concerned not so much with relating the diverse series [*Reihen*] to the atom, as rather with *relating them reciprocally to each other* through the mediation of the concept of the atom" (277/208).

The law of multiple proportions expresses a regularity in the macroscopic combining weights of elements, considered pair by pair. The atomic hypothesis introduces a further demand: each element must be assigned a characteristic 'atomic weight' that applies consistently across its compounds. This demand prompted chemists to extend their investigations, drawing upon further empirical laws governing different domains of chemical and physical phenomena. As this determination progresses, the range of empirical relations itself expands accordingly: "If we conceive this progress completed, then in the 'absolute' atomic weights all possible *relations* would be expressed, that the particular series could enter into among themselves", or, more precisely, in the absolute values of the *relative* atomic weights (277/209). On the one hand, by attributing observations concerning vapor density, heat capacity, isomorphism, and so forth to one and the same subject, we organize disparate known empirical laws into a system. On the other hand, in the attempt to determine the atomic weights, we discover previously unknown empirical laws. Cassirer can thus conclude with a characteristic neo-Kantian move: the atom should not be conceived as the 'substantial unity' from which scientific investigation begins, but rather as the 'projected unity' toward which it strives: "Thus the atom of chemistry is an 'Idea,' in the strict meaning Kant gave this term" (280/210).

1.3 From Substitution to Valence

From a logical point of view, one can say that the first stage in the development of chemistry was the ability to determine the relative atomic weights of *individual* elements. In Cassirer's reconstruction, the second stage was arranging them into *classes*. Following his sources, Cassirer points out that "[t]he empirical facts leading to such relative distinctions and combinations within the total system are given in the relations of chemical *substitution*" (281/211). It was Jean-Baptiste Dumas who noted that, when acetic acid is treated with chlorine, for every volume of hydrogen removed, an equal volume of chlorine is taken up, producing trichloroacetic acid, which reacts in almost the same way as the original compound. Examining how different elements substitute for one another reveals basic rules governing these substitutions: "The form of substitution can be determined once for all and expressed by certain *numerical values*, which we attach to each element along with the numerical value of its combination weight" (281/211).

To construct a systematic account of chemical substitution, chemists needed a common unit for comparing the combining capacities of different elements. Just as hydrogen served as the conventional standard for relative atomic weights, its combining capacity could serve as the unit of valence. Within the classical valence scheme, chlorine, oxygen, nitrogen, and carbon were assigned combining capacities of one, two, three, and four, respectively: Cl $\sim$ 1H, O $\sim$ 2H, N $\sim$ 3H, and C $\sim$ 4H. This establishes a

further basis for relating the individual elements to one another, introducing a "new characteristic constant" for each elementary substance: "The 'valency' of the elements is the expression of a definite property in them, that belongs to them independently of their chemical affinity" (Cassirer 1910, 281/211). The concept of valence arises from the demand for consistency. Without it, one might merely record individual cases in which one element replaces another in compounds. Valence brings them under the general rule that substitution must preserve the total valence, allowing one to determine a coherent system of substitution relations across different compounds.

Arranged according to this rule, chemical compounds fall into various 'principal types' (*Haupttypen*) (281/211–12). Compounds belonging to the same type can be derived from a common basic form through successive substitutions governed by the valences of the elements involved. Substitution thus becomes an instrument of classification. Methane gas provides a simple illustration. In this bulk substance, carbon exhibits a total combining capacity of four, which is entirely satisfied by a proportional mass of hydrogen. Because chlorine has the same valence as hydrogen, one can progressively react methane with chlorine to generate a series of new substances. Each successive substitution strictly preserves carbon's overall fourfold requirement:

$$\mathrm{CH_4} \longrightarrow \mathrm{CH_3Cl} \longrightarrow \mathrm{CH_2Cl_2} \longrightarrow \mathrm{CHCl_3} \longrightarrow \mathrm{CCl_4}.$$

Although these compounds exhibit markedly different physical properties, they are understood chemically as modifications of the same fundamental 'methane type,' because each step involves exchanging exactly one equivalent of hydrogen for one equivalent of chlorine. The same classificatory principle is illustrated by the water type. Despite their different properties, water, alcohols, and ethers share the same basic pattern of two attachments to oxygen.

Following the logical rather than the historical order, Cassirer could thus regard the concept of valence as a foundation for the concept of 'chemical type.' What interests him is that chemical 'substitution' illustrates a form of classification in chemistry fundamentally different from the one prevailing in the descriptive sciences. Whereas traditional classification groups substances according to properties they share, the chemical type establishes a systematic relation among compounds through rules governing their possible transformations:

> The concept of the 'type' here comes in for consideration not in its significance for the special problems of chemistry, but as a paradigm of certain *logical* relations. In fact, it displays most distinctly a characteristic feature already established in general by the analysis of the exact scientific concept. The chemical concept of type is also not formed on the pattern of the genus-concept, but on that of the sequence-concept. The different combinations, belonging under a type, are not so conceived because of the external similarity of their sensuous properties, or because of the direct agreement in their chemical functions. They belong together in so far as they can be changed into each other by means of the relation, which subsists between the valency of the individual atoms, while the remote members of the series need no further *analogy* than is established by this law of derivation itself. (282/212[16])

Dumas still grouped compounds derived from one another by *analogy*, insofar as they retained similar chemical properties; Regnault extended the notion of *substitution* to encompass a whole sequence of reactions yielding products whose properties differed completely from those of the initial compound. The unity of the chemical type rests

[16]Translation modified.

not on the similarity of the actual substances themselves, but on the rules, grounded in 'valence,' that govern their possible transformations through substitution. Traditional *conceptual abstraction* establishes the unity of a class by identifying properties shared by its members. Chemistry mimics *mathematical abstraction*, which, by contrast, can establish conceptual unity through relations preserved under specified transformations. In projective geometry, two figures, however different in appearance, may belong to the same *equivalence class* if one can be transformed into the other by a collinearity-preserving transformation. Similarly, two chemical compounds, however different in their properties, may belong to the same *chemical type* if one can be derived from the other through a succession of valence-preserving substitutions.

One might be tempted to explain valence as a "thingly quality", as if each atom possessed a fixed number of 'hooks' (Cassirer 1910, 284/214). Cassirer, however, dismisses these mechanical models as *obscurum per obscurius* explanations (285/214). The real advantage of attributing to each atom its own valence is that it imposes a demand for consistency (283/213). The different valences must be accounted for within a coherent scheme, rather than assigned independently to accommodate each compound. At the empirical level, different combining capacities can be accepted as separate facts. The atomic hypothesis prevents chemistry from accepting this inconsistency and forces it to organize experience into a coherent system of chemical types: "what distinguished the concept of valency from all scholastic qualities is the intellectual renunciation, which it implies. It does not seek to penetrate into the substantial nature of the connection of atom and atom, but merely to represent the facts of this connection according to universal quantitative principles of order" (285/214).

Cassirer concludes that the implications of this development become particularly evident in organic chemistry, where the theory of substitution acquired greater scope once chemists recognized that groups of atoms (*Atomgruppen*), as well as individual atoms, could function as units of substitution. Although these concepts emerged historically in a different order, Cassirer appears primarily interested in their logical dependence. The conversion of methanol into chloromethane,

$$\mathrm{CH_3OH + HCl \longrightarrow CH_3Cl + H_2O\,},$$

illustrates how a group of atoms can be treated as remaining unchanged through a chemical transformation. Here, $\mathrm{CH_3-}$ functions as a 'compound radical' (*zusammengesetztes Radikal*), while OH is replaced by Cl. Radicals could replace one or more hydrogen atoms in a compound and were accordingly classified as monovalent, divalent, and so on. For example, methyl ($\mathrm{CH_3-}$) has one point of attachment, whereas methylene ($\mathrm{CH_2}$) has two. Whereas Laurent conceived of radicals as 'nuclei' persisting within compounds, Gerhardt treated them as 'residues,' emphasizing their role in relating different compounds through substitution (286/215). Once again, Cassirer can argue that this alternative (Ladenburg 1902, 175) rests on a mistaken assumption. In the expression $\mathrm{CH_3-}$, $\mathrm{CH_3}$ designates the methyl radical, while the free valence indicates the possible substitutions. For Cassirer, the significance of 'radicals' therefore lies not in their correspondence to actual, independently existing, thing-like entities, but in their function of constraining the range of allowable substitutions. In a somewhat contrived way, Cassirer concludes that here chemistry seems to set aside "the questions as to whether and how the elements continue to exist in the compounds into which they enter" and seeks only to discover the measurable relations "between the initial and final conditions of a process of chemical transformation" (Cassirer 1910, 286–87/215–16).

2 The Polemic with Rickert. The Case of Isomerism

Like the history of physics, the history of chemistry can be understood as a process of progressive 'arithmetization,' in which qualitative differences are replaced by purely numerical differences (Cassirer 1910, 289/217). Indeed, Cassirer closes the chemistry section with a detailed reconstruction of Lothar Meyer's establishment of the *periodic system*, which arranged the individual elements by increasing atomic weight and grouped them by valence (288–92/216–220). Yet the section on chemistry was not merely an appendix to the previous sections on physics, as Cassirer points out at the opening of section IX: "The real *methodological* interest of the construction of the concepts of chemistry lies in the fact that in it the *relation of the universal to the particular* is set in a new light" (292/220). Considered in isolation, physics seems to give rise to a traditional antinomy: 'knowledge' aims at the universal in the form of general laws, whereas genuine 'being' belongs to individual things. Chemistry, by contrast, seems from the outset concerned with determining the distinctive properties of individual substances (*Einzelding*), such as gold, water, or particular organic compounds.

Cassirer's aim in his discussion of chemistry was to show that these two directions of concept formation in the natural sciences are mutually compatible and, indeed, inseparable. It is worth noting that the expression 'concept-formation' (*Begriffsbildung*) does not play any role in the other works of the Marburg school. For example, the term does not appear at all in Natorp's contemporaneous 1910 book. By contrast, as is well known, the term was given prominence by the competing Heidelberg–Freiburg school. In particular, Heinrich Rickert's (1902) work had initiated a debate in contemporary German philosophy concerning the 'limits of concept formation' in the natural sciences in relation to the cultural sciences. The more universal a scientific law or concept becomes, he claims, the further it moves from actual reality. To grasp reality in its individuality, human knowledge must adopt a different, complementary methodology: the individualizing method of the historical and cultural sciences. By framing his work in terms of the debate over scientific concept formation, Cassirer clearly sought to expose the inadequacy of Rickert's sharp distinction between the *generality* of the natural sciences and the *individuality* of the historical sciences. However, it was only after treating chemistry that Cassirer was in a position to launch a direct attack on Rickert, thereby concluding the first part of the book.

As one might expect, for Cassirer Rickert's conclusion is the consequence of a misunderstanding of the nature of concept formation in the mathematical sciences. Here, generality is not achieved through the progressive omission of individual differences, but through the transformation of individual differences into numerical variables. Initially, we might describe, say, a volume of gas in terms of its perceptible characteristics. Referring to Duhem (1906), Cassirer points out that physics first replaces the real gas with an idealized model whose behavior is characterized by only a few variables (pressure p, volume v, and absolute temperature T) (Cassirer 1910, 189–90/143–44), connected by a functional relation, the ideal gas law:

$$pv = RT\,.$$

The ideal gas is an abstract representation of a real physical system: it describes how a gas *would* behave under idealized conditions in which additional physical effects are neglected. At this point, Rickert's objection might still seem justified. In this model, all gases are treated as equivalent: their individual characteristics are disregarded, and

their behavior is represented by the same general functional relationship (Cassirer 1910, 304/228). It seems that, to recover more specific detail, we must sacrifice the generality of the law.

According to Cassirer, however, concept formation in physics proceeds in precisely the opposite direction. To account for the differences between 'real' gases and the 'ideal' gas, physics moves to a more 'general' functional relation that encompasses the ideal gas law as a special case. The van der Waals equation,

$$\left(p + \frac{a}{v^2}\right)(v - b) = RT\,,$$

introduces two additional empirically determinable 'degrees of freedom' , a and b, whose values differ from one gaseous substance to another (see (Cassirer 1936, 107–08/85–86)). The ideal gas law is recovered by setting $a = b = 0$. The more general mathematical framework thereby permits a more differentiated empirical characterization of particular gaseous substances: "it appears that the scientific concept is in no way denied the establishment of the individual" (Cassirer 1910, 306–7/231). The movement toward greater systematic generality is simultaneously a movement toward the more precise determination of the particular.

Rickert might still object that the goal of physics is the search for general laws, and not the determination of individuality. According to Cassirer, however, this objection loses much of its force once we move from the *laws* of physics to the *formulas* of chemistry. Indeed, the goal of the formula is precisely to identify an individual compound and its properties, as distinct from those of other compounds. Nevertheless, here too, chemistry must turn to a more general determination to grasp more particular cases. To establish this point, Cassirer appeals to the phenomenon of 'isomerism' . Before the recognition of isomerism, chemists generally assumed that when two distinct substances share the exact same empirical or *compositional* formula

$$\mathrm{C_2H_6O}\,,$$

that is, they are 'indistinguishable' at the level of elemental percentage composition. If chemistry followed the traditional logic of concept formation, grasping the individual difference between the two isomers would require adding more *concrete* sensory descriptions (color, boiling point, odor). Chemistry, however, proceeds in the opposite direction, introducing more *abstract*, two-dimensional *constitutional* or structural formulas

$$\mathrm{CH_3{-}O{-}CH_3} \qquad \mathrm{CH_3{-}CH_2{-}OH}$$

which specify how the atoms are bound together through valence units to form the molecular framework (276/207).

> Thus if the chemical "concept" of a certain body is given by its *constitution-formula*, in which it is grasped as a particular material in its characteristic structure, it is at the same time brought under the various chemical "types," and is thus set in a definite relation to the totality of remaining bodies. The ordinary chemical formula only gives us the composition in general, but not the type of construction of the individual elements; here it is enriched by a wealth of new relations. The general rule now in our possession enables us to trace how and according to what law the given material is transformed into another; the rule involves not only the form of the existence of the material at a definite moment, but the totality of its possible spatial and temporal phases. The further the chemical construction of concepts proceeds, the more sharply the particulars can be *distinguished*. Materials, which were called similar, because "isomeric," from the

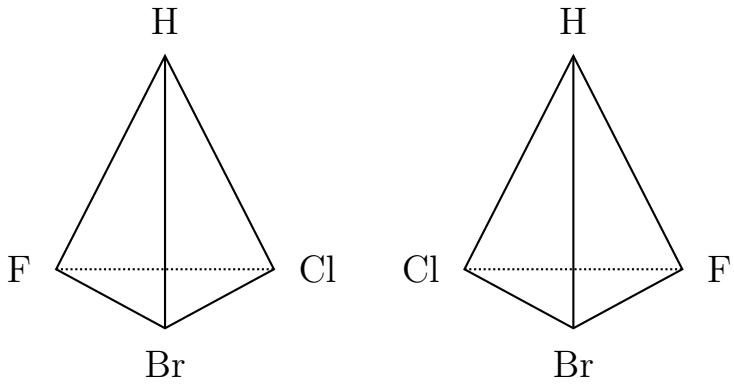


Figure 1: Enantiomers of fluorochlorobromomethanes; redrawn after Duhem 1902, 139

> standpoint of the undeveloped concept, are clearly separated and distinctly defined in character, from the standpoint of the developed concept. (Cassirer 1910, 298–99/224)

But the process did not end there. Compounds were found, the so-called enantiomers, that share the same structural formula while exhibiting opposite optical behavior. Optical isomerism thus revealed the limitations of two-dimensional structural formulas, just as constitutional isomerism had exposed the limitations of compositional formulas. Once again, the *individual* differences between substances could be captured by introducing more *general* forms of representation. Van 't Hoff replaced the two-dimensional structural formulas with three-dimensional stereochemical formulas (fig. 1). Of course, Cassirer resisted the idea that these formulas depict the *actual* spatial arrangement of atoms; rather, they encode all *possible* reactions that the compound might undergo under different conditions: "Since the constitution-formula represents this *connection*, this formula is indeed the real scientific expression of the *empirical* reality of the body; for it means nothing else than the thorough-going objective connection, in which an individual 'thing' or particular event stands with the totality of real and possible experiences" (285/214).

Rickert might object that an individual chemical compound is a repeatable chemical kind rather than an individual in the strict sense. However, this objection would amount to conceding that his real concern is not that scientific concepts fail to capture the 'individuality' of objects (*Einzelheit*), but that they fail to capture their concrete 'thinginess' (*Dinglichkeit*) (305/229). These, however, are distinct problems. What counts as an 'individual' is not given to theory as an external 'thing,' but is determined within the theory itself by its own 'general' concepts. Different forms of inquiry—chemistry, biology, history—employ different criteria for individuating their objects. In this sense, and only in this sense, the integration of these different forms of 'individuality' can never be completed. As one might expect from Cassirer, it remains an infinite task: "Here, in fact, the 'individual' reality shows its fundamental character of inexhaustibility. But it is the characteristic merit of the true scientific relational concepts, that they attempt this task in spite of the impossibility in principle of its completion" (309/232).

3 A Battle over the Substance-Concept. The Meyerson–Cassirer Debate

Just after the book was completed in July 1910, Cassirer asked his publisher to send Meyerson part of the galley proofs of *Substanzbegriff und Funktionsbegriff* that

contained some short remarks on Meyerson's philosophy (Cassirer 1910, 430–431/324). Cassirer hoped that they could continue their "battle over the substance-concept" (Cassirer to Meyerson, Aug. 27, 1910[17]) on a level playing field before Meyerson finished the review. However, Meyerson had already drafted it. By the time he received the complete proofs, it was already in the course of publication: "I am now in possession of your *Substanzbegriff* in its entirety, albeit without the cover page. This is, however, unessential, since I shall in any case have the book bound. The (very long) article on your *Erkenntnisproblem* has for some time been lying corrected at the editorial office, but, as I was recently informed, it may not yet appear in the next issue, for typographical reasons!" (Meyerson to Cassirer, Oct. 21, 1910[18]).

3.1 Meyerson's Review of Das Erkenntnisproblem

The nearly thirty-page review appeared in January 1911. Surveying both volumes of *Das Erkenntnisproblem*, Meyerson praises the book for having integrated history and philosophy of science, a combination that is especially rare in Germany (Meyerson 1911, 101). However, he identifies a glaring "sin of omission" in Cassirer's historical reconstruction: "The author has not, it seems to us, paid sufficient attention to the history of chemistry" (102). This omission is especially striking in a German book, given that German scholarship had produced Hermann Kopp's (1843–1847) monumental four-volume history of chemistry. For Meyerson, Cassirer's 'sin of omission' is not a mere oversight; it reflects a tendency to downplay evidence that might challenge his functionalist conception of scientific knowledge. Meyerson concedes that Cassirer might argue that, until the second half of the eighteenth century, chemistry exerted much less influence on philosophy than physics did. The sole exception would be Boyle, both a chemist and a physicist, whose work Cassirer had indeed discussed extensively (Meyerson 1911, 102).

However, in Meyerson's view, Cassirer's treatment of Boyle in the second volume of *Das Erkenntnisproblem* is, at the very least, tendentious. Cassirer considers Boyle's contribution to epistemology solely in terms of his having replaced the Renaissance organicist conception of nature with a conception of nature as a "system of rules" (Cassirer 1907, 312–313). For Meyerson, this reading overlooks the fact that Boyle was not merely an advocate of a lawlike conception of nature; he was "a rigid atomist, assuredly one of the most rigid that the history of science has ever known" (Meyerson 1911, 122–123). Had Cassirer examined Boyle's work from this perspective, Meyerson argues, he would have recognized the indispensable role not only of functional *legality* but also of the concept of *substance*. This conclusion can be avoided only by resorting to the familiar positivist "subterfuge" of claiming that Boyle proceeded *as if* atoms existed (123). To be sure, Cassirer does not resort to this "rather crude device"; instead, he adopts the strategy of Marburg idealism, in which the specific form of physical hypotheses is justified by the application of the "concept of function" (123). For Meyerson, however, the result is ultimately the same.

Meyerson takes the omission of Lavoisier's work from *Das Erkenntnisproblem* as further evidence of Cassirer's disregard for the history of chemistry (Meyerson 1911, 103). If Boyle could be included in the book as a physicist, treating Lavoisier would inevitably have forced Cassirer to address the entire development of chemistry, to which Lavoisier's doctrines remain "indissolubly linked" (103). For Cassirer, the development of science

[17] ECN, Vol. 18, Doc. 153.
[18] Vol. 18, Doc. 157.

involves a progressive replacement of 'things' by a mathematical system of functional 'relations.' However, For Meyerson, however, Cassirer's account is historically untenable. When science destroys a 'thing,' "it immediately creates another, or rather it creates before destroying, for it can destroy only by creating" (129). The fate of phlogiston illustrates the point. Its rejection did not reduce chemical reality to mathematical functions; it was accompanied by the positing of new entities, such as oxygen, to explain the phenomena: "The concept of a purely phenomenal reality, of a world of relations without a support, is entirely foreign to science; science imperatively demands a world of objective realities" (129).

3.2 Chemistry in the Second Edition of Das Erkenntnisproblem

Had Meyerson been familiar with *Substanzbegriff und Funktionsbegriff* when he reviewed *Das Erkenntnisproblem*, he would presumably have tempered some of his critical remarks. As we have seen, Cassirer not only devoted an entire section to concept formation in chemistry, alongside his discussions of physics, but also attributed to chemistry a distinctive "methodological interest," directly contradicting what Meyerson seems to suggest (Cassirer 1910, 292/220). From Cassirer's perspective, Meyerson's philosophical position was not a consequence of his historical investigations into chemistry. Rather, Cassirer could have turned Meyerson's accusation against him, arguing that Meyerson's interpretation of the history of chemistry was itself shaped by a mistaken conception of scientific concept formation. If Cassirer had neglected Boyle and Lavoisier, Meyerson, Cassirer might have objected, was equally guilty of downplaying the contributions of Richter and Dalton. Nevertheless, Cassirer took Meyerson's criticism seriously: "Of course, you already know that I cannot agree with you on everything, but your critique contains such a sharp and concise formulation of the contrast between us that it already contributes significantly to a mutual understanding" (Cassirer to Meyerson, Feb. 21, 1911[19]).

At the time Cassirer wrote this letter, the second edition of the first volume of *Das Erkenntnisproblem* had already been published, but Cassirer promised to take Meyerson's comments into account in the second volume. After visiting Meyerson in Paris at the end of March 1911 (Cassirer to Meyerson, Mar. 30, 1911[20]), Cassirer wrote to him again just after the second volume of *Das Erkenntnisproblem* appeared: "You will notice for yourself that I have made extensive use of your review, and that it has been extremely valuable to me: for example, I have now added a new section on the development of chemistry[21]" (Cassirer to Meyerson, Oct. 25, 1911[22]). In the new section, devoted entirely to the problem of matter in chemistry, Cassirer makes a concerted effort to atone for his 'sin of omission' by addressing the historical dimension of Meyerson's objections. At the same time, however, he skillfully integrates these historical details into his own conceptual framework.

Cassirer now discusses the role of Boyle's corpuscularism in the history of chemistry, but presents it as the first step in a process in which nature is stripped of its "inner qualitative strivings and forces", and Boyle conceives of nature not as a separate agent but as a "system of rules" prescribing how corpuscles act upon and are acted upon by one another (Cassirer 1911, 2:433). Still, Cassirer suggests that the influence of this

[19] Vol. 18, Doc. 161.
[20] Vol. 18, Doc. 164.
[21] Cassirer 1911, 2:428–441.
[22] ECN, Vol. 18, Doc. 178.

view on chemistry proper was initially meager. John Freind (1709) sought to reconstruct the entire discipline along Newtonian lines, introducing the concept of attraction as an explanatory principle. Such attempts, however, ultimately remained sterile, since they were still insufficiently supported by experience (Cassirer 1911, 2:434). The first half of the eighteenth century remained dominated by the phlogiston theory, in which "the qualitative conception of natural processes once again comes characteristically to the fore" (2:434).

Cassirer now acknowledges that "only with Lavoisier's reform [...] is the scientific concept of chemistry itself transformed" (2:435). Whereas *Substanzbegriff und Funktionsbegriff* mentions Lavoisier only in passing, the second edition of *Das Erkenntnisproblem* places him at the center of the narrative. While the phlogistonists treated quantitative measurements as secondary, Lavoisier made the quantitative relations between the initial and final states of chemical transformations the criterion for identifying a 'substance.' Once again, however, Cassirer makes this case study work in his favor. Whereas Meyerson emphasized the continuity of explanation in terms of persistent constituents across the transition from phlogiston theory to Lavoisier's chemistry, Cassirer interpreted the same episode as a functional reinterpretation of the concept of substance: "for here the concept of substance passes definitively from its ontological into its modern epistemological-critical meaning, being conceived as a pure correlative concept to the idea of quantitative conservation" (2:440).

The Meyerson–Cassirer 'battle over the substance-concept' was fought with historical weapons, but the battlefield was ultimately a purely systematic one. Cassirer was ready to accommodate Meyerson's objections on historical grounds, though he entirely rejected their systematic implications. Meyerson contends that scientists continuously appeal to 'things' that remain constant beneath the apparent variety of phenomena (Boyle treats his corpuscles as being as real as tables and chairs). By contrast, Cassirer could easily reinterpret the same episode as confirming his narrative of the progressive reduction of the substance-concept to the function-concept (Boyle's corpuscles move according to mathematical laws). For all their differences, Meyerson found the second edition of *Das Erkenntnisproblem* "considerably improved" (Meyerson to Cassirer, Nov. 15, 1911[23]). Eager to see his work gain recognition, Meyerson likely appreciated that Cassirer, a rising figure in German philosophy, had taken his suggestions seriously, even if he incorporated them into a very different narrative. If the Meyerson–Cassirer debate teaches us anything, it is perhaps the limitations of such a *grand récit*: both Meyerson's and Cassirer's narratives turn out to be at once too narrow to accommodate the full range of historical evidence and so broad that virtually any case can be made to fit them.

Conclusion

In December 1912, while announcing the publication of the second edition of *Identité et réalité* (Meyerson 1912), which included numerous references to Cassirer's works, Meyerson sent Cassirer two issues of the official organ of the French Philosophical Society devoted to a discussion of Perrin's study of Brownian motion:

> At the same time, allow me to send you two issues of the Bulletin de la *Société française de philosophie*, which you probably do not see regularly. They contain Perrin's lecture on the new measurements of atoms, as well as the ensuing discussion. I also took part in

[23] ECN, Vol. 18, Doc. 180.

> the discussion and, of course, I attempted to turn the situation to my own advantage. Although you will probably disagree with almost everything, you might perhaps be interested in certain aspects of the whole subject. Should you write anything about it (for which I would be very grateful), I kindly ask you to send me a copy. (Meyerson to Cassirer, Dec. 1, 1912[24])

Because chemical experiments could only measure *relative* atomic weights, some nineteenth-century chemists could treat atoms as a useful mathematical fiction. Using independent measurements (Brownian motion, electrical measurements, and radioactivity), Perrin was able to determine Avogadro's constant (Perrin 1910, 281), providing evidence for the existence of atoms with determinate *absolute* weights in grams. During the discussion, Meyerson did indeed turn this result to his 'own advantage.' Although physicists might often be 'professed positivists,' they were invariably 'practicing realists' who could not resist treating atoms as 'real' entities. By sending the paper to Cassirer, Meyerson wanted to 'score a point' in their ongoing debate. Cassirer's response to Meyerson's letter has a cordial yet detached tone (Cassirer to Meyerson, Jun. 23, 1913[25]). Indeed, he might have felt that Meyerson's objection rested on a fundamental misunderstanding of the issue at stake.

If we consider Cassirer's relatively sparse remarks on the subject in his epistemological writings over the following decade, it becomes clear that his concern was not with the existence of 'atoms,' a question he left to science to decide, but rather with the nature of the scientific 'atom-concept' (*Atombegriff*). By following the history of the atom-concept, according to Cassirer, one can see how the constancy of the atom as a thing (*Dingbegriff*) gives way to the constancy of a numerical ratio (*Maßbegriff*): "fundamentally, such a realization, such a strictly quantitative meaning of the concept of the atom, was only reached in the beginnings of modern chemistry in the law of multiple proportion" (Cassirer 1921, 17; tr. 1923, 360). As we have seen, chemistry progressively determined *relative* atomic weights through the convergence of independent lines of evidence; Perrin's work subsequently enabled the unambiguous determination of the *absolute* mass of an individual atom through the convergence of independent measurements. For Cassirer, the logical significance of these measurements lies not in the proof of the existence of atoms, but in the fact that the assumption of the existence of atoms leads to the determination of a universal constant. The fact that heterogeneous phenomena can be coordinated through a single numerical constant provides the only warrant for the *objectivity* of physical knowledge, understood as the progressive 'de-anthropomorphization' of nature: "Physical thought is always concerned at first with establishing a characteristic point of view of measurement in an objective physical concept, in a determinate constant of nature" (Cassirer 1921, 17; tr. 1923, 359[26]).

Cassirer appropriated this 'Planckian' move, replacing 'realism' about theoretical entities with the 'objectivity' of measured numerical constants, in various forms throughout his subsequent writings (Cassirer 1929, 520/446). However, it was only in Cassirer's (1936) later monograph on the new quantum theory that he found the opportunity to organize these remarks into a coherent form. The question of the reality of atoms has always led to the opposition between 'positivism' and 'realism.' However, this is a false dichotomy. Those who regard atoms as 'fictions' cannot account for the

[24] ECN, Vol. 18, Doc. 195.
[25] Vol. 18, Doc. 203.
[26] Translation modified.

extraordinary heuristic power of the atomic hypothesis; those who regard them as 'things' fail to account for the constant change of atomic images. This dialectic can be overcome once one recognizes that the ultimate value of the atomic hypothesis lies in its *Spürkraft*, its capacity to uncover previously unknown constants of nature. For example, the kinetic theory of gases leads to the determination of Loschmidt's constant: "These are new universal constants to whose knowledge the atomic hypothesis leads—knowledge possessing the *highest degree of 'objectivity' that natural science can possibly achieve*" (178/143[27]). The model-like representations of the atom may change while the universal quantitative relations they establish survive (179/144).

Once again, Cassirer resorts to a lengthy historical reconstruction of the development of the 'atom-concept' to make his point. 'Physical atoms' were initially introduced speculatively as small, impenetrable, 'hard' bodies. However, as Lasswitz (1890) showed, Huygens introduced atoms as a consequence of the conservation of energy and momentum. While Boyle conceived of 'chemical atoms' as thing-like entities, Dalton presented them as a by-product of the empirically confirmed law of multiple proportions (Cassirer 1936, 183/147):

> As with Huygens, the atom is here defined strictly substantially; but all statements concerning its nature [*Was*] ultimately merge into statements concerning quantity-definite numerical relations [*Wieviel*]. All the basic concepts and theories of chemistry, the concepts of valence and types, the concept of the radical as well as the theory of compound radicals all point in this direction. The conceptual language of atomism in conjunction with the general symbolic language of mathematics now permits that simple and magnificent development in the language of chemistry that is evident in the progress from simple to structural formulas. Stereochemistry, as established by van 't Hoff, permitted the elucidation, through a complex construction of formulas, of an entirely new group of facts, the phenomena of stereoisomerism; it threw new light on the distribution of atoms in space. Thus the visual picture of the atom is modified and refined, but all these changes serve only the one purpose of circumscribing and representing an increasingly large domain of functional relations. (183–84/147)

The discovery of the electron brought together the previously separate histories of the chemical and the physical atom. The indivisible chemical atom turned into a composite system of electrons (Thomson's 'plum pudding' model) and, after the discovery of the nucleus, into a spatially structured system (Bohr's planetary model) (184–186/148–150). From this point of view, as Cassirer put it, it is hard to avoid the impression that the history of atomic theories offers a "kaleidoscopic" variety, a strange "phantasmagoria" of disconnected portraits (186/149). On closer inspection, however, each theory identifies and corrects the errors of its predecessor while advancing "in a single direction" (187/150). What matters is not what the atom ultimately 'is,' but the historical stability of the universal constants that it allows us to uncover: "As long as this empirical interconnection is safeguarded, and as long as we are convinced that the progression of 'atomic models' is guided not by mood or caprice but by a definite rule, the atom has all the 'reality' of which it is capable" (Cassirer 1936, 187/150).

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[27] Translation modified.

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